

Business process and risk models enrichment: considerations for business intelligence

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Abstract

Business Intelligence (BI) as a paradigm with methods and tools substantially improves decisions with regard to business evolutions. Through the emerging business process management approaches, it becomes possible to link BI-tools with the underlying process models. This paper explains how process models can be enriched with risk models in the context of business intelligence.

1. Introduction

Nowadays organizations are subjects to frequent changes requiring continuous alignment of business processes on business strategies. A business process is an asset for value creation [1], which is exposed to incidents that may even imply a business interruption. It is also a context for various interactions and a source of incidents. Thus, business processes are exposed to the same quality requirements as materials and human resources. Practitioners often handle processes from the perspective of goal alignment and performance improvement while neglecting the aspect of value preservation. However, an emerging research community on process governance and compliance [2, 3] suggests a global approach to risk management in organizations based on business processes.

Considering the complexity of business risks and the importance of mastering these, an integrated management method for risk and process is suggested in [2]. This paper extends the previous work with facilities for the monitoring of business risks with regard to performance. In the practice, the monitoring of business process performance through indicators is tailored in the context of Business Intelligence. Within process management, it is possible to link BI-tools with

the underlying process models for the implementation of business activity monitoring. Here, BI shall support the monitoring phase of the robust management process.

This paper explains how a method for integrated management of process-risk can be supported in the context of BI.

First, the Meta model for performance and risk indicators is introduced. Finally, an elaborated scenario is proposed before concluding.

2. On performance and risk indicators

Organizations exist to achieve missions, which are concretized through objectives. The later is what the “entity strives to achieve”. An objective is also a “statement of preference about possible and achievable future situations that influence the choices within some behavior.”[4].

In the practice, the achievement of an objective is quantified through performance indicators. The later is a metric, which provides information about the behavior of the performer. In fact, an indicator states how well an objective is being achieved given the resources. Also, approaches to enterprise risks management and business continuity management address business risk under the aspect of performance preservation. Here, the variation of process performance is highly coupled to the presence of a risk. Performance variation can be seen as the measure (standard deviation as stated in [5]) of the risk but also the indicator of the risk. The later are risk indicators [6].

From the risk perspective, indicators are measures that indicate the level of changes in an organization’s risk profile. According to [7], a risk indicator is a “predictive [measure] regarding changes in the risk

profile of a business... [and] enables timely action to be taken to deal with issues arising”.

An important objective of risk indicators is to provide a measure of changes in the probability of risk factors, root causes and to enable therefore reactivity through real time monitoring. In addition, indicators are useful for the identification and control of weaknesses and/or opportunities, since they provide guidance for analysis. In fact, as stated in [8] indicators may support the monitoring of the actual negative impact of risks. At the process level, indicators may support the monitoring of the quality of operational processes regarding risks.

The relations between performance indicator, risk indicator and business specific concepts are layout in Figure 1. Any business area or domain is defined in order to perform a set of objectives performed by business processes and enterprise activities. A performance indicator is associated to an objective through a quantification relation, which states that the

indicator is used to quantify the achievement of that objective. At this conceptual level, a parameterized equivalence relation between a performance indicator and a risk indicator shows to which extent the former provides information about the risk profile. The risk indicator is indeed a function of the performance indicator. The indication relation between a risk and a risk indicator states, which risk is measured by the given indicator. When aggregated, all indicators of a given risk provide information about the overall risk profile at a given time. Based upon these relations, it is possible to evaluate the overall risk indicator profile for each domain, process and activity with regard to each objective. A tool support with an information system shall enable the navigation between performance profile and risk profile providing therefore performance driven risk analysis with regard to performance preservation.

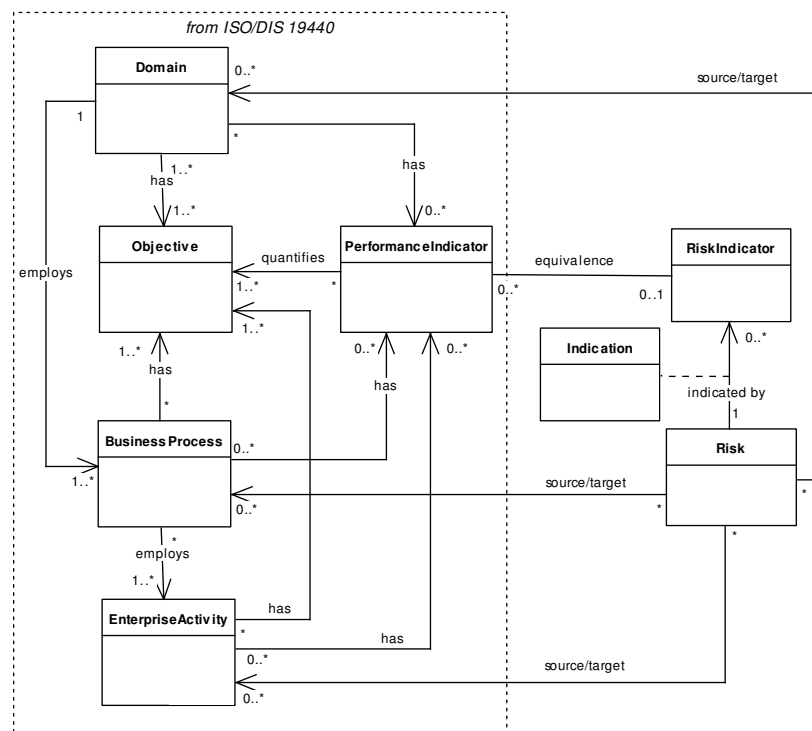


Figure 1: Meta model integrating performance and risk indicators

3. Illustration

The simple case study of Figure 2 (defined in the Even driven Process Chain formalism of the ARIS framework) shall illustrate how the extension supports

indicators based risk-driven analysis of activities. Consider the activity “assemble computer components”, which generates three events. Reliability and productivity are the main indicators explained for this activity. A disjunction between the resources “computer assembly manual” and “hardware

requirement” is a factor, which influences the likelihood to get stuck in the unwanted state “computer components not assembled”. Hereby, the achievement of the objectives “assembly schedule” and “target workload” is affected. Regarding this illustration, the possibility to get into the situation, where components are not assembled is a risk. Given the effect of this risk on the activity “assembly schedule”, it causes a decrease of the reliability. This situation is perceived through the variation of the reliability indicator. In this particular situation, the decrease of reliability greater

than 1% is considered to be a risk indicator. Figure 2 shows visually the relations between risk and risk indicator. The risk is modeled as a hexagon with an exclamation mark. A double arrow between a risk and an activity states that the risk is generated in the context of the activity. The relation between the performance indicator and the risk is represented as a dotted line. The label of this line describes the function, which transforms the performance indicator into the given risk indicator.

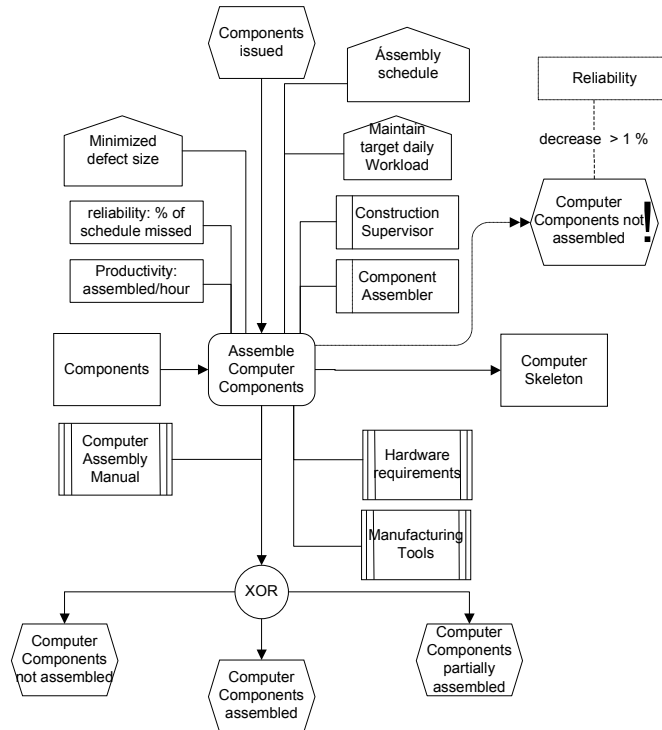


Figure 2: Sample activity with risk, risk indicator and performance indicator

4. Related work

The literature points out many approaches to the management of process risks. Risk centered approaches, which emphasize only risk management. Process oriented approaches, which consider risks from the information and simulation points of view. Industrial frameworks, which suggest the management of business process risks without providing tools for the analysis of the relations between risks and processes.

In contrast to these, structural approaches such as Bayesian networks and the fishbone diagrams focus on causalities. Others useful approaches are industrial events analysis such as Failure Modes and Effects

Analysis (FMEA), Fault Tree Analysis (FTA), Event Tree Analysis (ETA) also focus primarily on causal analysis.

In our approach, risk is considered as a complex concept with a well defined structure. A Meta model is defined and unified with a Meta model of enterprise modeling [2].

5. Conclusion and future work

Enhanced with risk management, a process management system shall be able to anticipate outcome uncertainties of business processes. In this paper, we studied the relations between risk and performance indicators while considering an example. Currently, we

further investigate on tools for supporting the analysis of the business process risks. In the context of BI, this includes the linkage between performance and risk indicators and the preservation and reuse of the specified risk model enhanced business process models. Scenarios will then be defined and evaluated in business cases

6. References

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